

V2: Directional Forcing

9 October 2026 (rerun)

When does a second wave direction break the single-source inverse?

Amir Dollie (DLLAMI004), EEE4022S

SHARC Code: `runDirectionalForcingStudy.m`, `v2Tools.m`, `directionalSpectralMatrix.m`, `diagnoseV2CoherenceNoise.m`,
commit 2880cd9

Results: `Results/V2/V2_20261009_133308`, `diagnostics/V2noise_20261009_134221`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 5 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). Settings unchanged; tester 13/13. This summary supersedes V2_20261005_142206. The bias significance per unit secondary energy rose three- to fourfold. The coherence warning improved more, about fifteenfold, because the higher signal-to-noise ratio raises the single-direction coherence from about 0.99 to 0.999. The early warning now holds at 30° as well, and is more robust to noise-floor error.

Summary

Question. The output-only inverse rests on one incident wave component. Then the spectral matrix is rank one and $T_j = S_{jr}/S_{rr} = H_j/H_r$. A second, independent direction makes it rank two, and the measured ratio becomes a mixture that the frozen inverse reads as a change of β and R . How large is that bias, and does the coherence loss, which is observable, warn before it matters?

Outcome.

- **The inverse is extremely sensitive to secondary energy.** All 16 registered cases ($\phi = 15$ to 90° , $f_2 = 5$ to 50%) are Not identifiable, with $d_{\text{sys}} = 30$ to 5559.
- **The tolerances are tiny.** $d_{\text{sys}} = 0.5$ is reached at **0.082%** of the incident energy at 15° separation, falling to **0.0057%** at 90° . A 1% stiffness bias needs 40 to 100 times more energy (0.47 to 3.3%).
- **The regime is linear.** This was verified directly and by a nonlinear inversion at $d_{\text{sys}} = 1$.
- **Coherence warns first for separations of 30° and more,** by factors of 2.6 to more than 11. It does not at 15° , but only narrowly (detection at 0.118% against 0.082%). The warning survives a ± 5 to 10% error in the assumed noise floor. Underestimating the floor by 5.8% produces false alarms.

Directional forcing is a structural model mismatch, not an estimator failure.

1 What the study does

Why V2 exists. Every earlier study (and the G3 inverse) assumes one coherent forcing direction. V2 violates that assumption on purpose while the inverse stays frozen, to find how much directional contamination it tolerates.

Why one direction matters. With one component, $\mathbf{y} = \mathbf{H}_1 u_1$, so $S_{jr} = H_{1j} H_{1r}^* S_1$ and $S_{rr} = |H_{1r}|^2 S_1$. Then $S_{jr}/S_{rr} = H_{1j}/H_{1r}$ exactly: the input spectrum cancels. With two independent components, $\mathbf{y} = \mathbf{H}_1 u_1 + \mathbf{H}_2 u_2$, and

$$\frac{S_{jr}}{S_{rr}} = \frac{w_1 T_{1j} + w_2 T_{2j}}{w_1 + w_2}, \quad w_c = |H_{cr}|^2 S_c.$$

This is a mixture of the two directions' transmissibilities, weighted by the *reference sensor's response energy* to each, not by the incident energy fractions alone. The signal spectral matrix is rank two unless the two response vectors are collinear, and no choice of (β, R) in the single-direction model reproduces it. The inverse can therefore converge perfectly and still be biased.

2 Method

Pre-registered in `cfg.v2`; a clarification and one amendment were logged before the first V2 run. Unchanged for the rerun.

1. **Forcing.** The primary component is at 0° (the EMM incidence, the Level 2C axis) and the secondary at $+\phi$, both with the reference JONSWAP shape. The total energy is fixed, $S_{u1} = (1 - f_2)S_u$ and $S_{u2} = f_2 S_u$, so directionality is not confounded with SNR. For a circular floe a component from heading ϕ is the nominal field at the sensors rotated by $-\phi$, so one EMM field at the rotated Level 2C points serves every case. Sweep: $\phi \in \{15, 30, 60, 90\}^\circ$ and $f_2 \in \{0.05, 0.10, 0.25, 0.50\}$, at p_0 , Level 2C, reference s26, LSM6DSV16X.
2. **Spectral matrix.** $\mathbf{S} = \sum_c \mathbf{H}_c \mathbf{H}_c^H S_{u,c} + \text{diag}(S_{nn,j})$, smoothed with the engine's Welch kernel. One shared function, `directionalSpectralMatrix`, is also used by D3. Both the expected transmissibility and its covariance come from this rank-two matrix through the unchanged engine functions.
3. **Bias.** $b_{\text{dir}} = f_W^{\text{dir}} - f_W^{\text{single}}$, both Welch expectations through the same path, projected with the nominal J , Σ and F :

$$\delta\theta_{\text{dir}} = F^{-1} J^T \Sigma^{-1} b_{\text{dir}}, \quad d_{\text{sys}} = \sqrt{\delta\theta_{\text{dir}}^T F \delta\theta_{\text{dir}}}.$$

The class uses the vector sum $\delta\theta_W + \delta\theta_{\text{dir}}$ under the nominal σ .

4. **Crossings.** Per ϕ , the first f_2 at which $d_{\text{sys}} = 0.5$, $d_{\text{sys}} = 1$ and $|\delta\beta/\beta| = 1\%$ are reached, by bisection in $\log_{10} f_2$ inside the bracket between the exact $f_2 = 0$ baseline and the first registered point.
5. **Coherence.** $\gamma_{jr}^2 = |S_{jr}|^2 / (S_{jj} S_{rr})$ per pair, with Fisher $z = \text{atanh}|\gamma|$. The expected drop of the band-mean z from the single-direction baseline is divided by its null scatter, *measured* by Monte Carlo of the real estimator (400

single-direction records). It is detectable when $\max_j D_j \geq z_{1-0.05/3} + z_{0.80} = 2.97$, the 80%-power threshold (the 5% rejection threshold is 2.13). Coherence warns early at ϕ if $f_{\text{detect}} < f_2(d_{\text{sys}} = 0.5)$.

6. **Checks.** Tester 13/13: the single-direction path reproduces the engine (f_W to 10^{-14}); $\pm\phi$ mirror symmetry on the EMM; rank and coherence; energy conservation; local linearity. A nonlinear G3 inversion is run at the most sensitive $d_{\text{sys}} = 1$ crossing. Run time 366 s.

3 Results

3.1 The registered 5 to 50% grid

ϕ	d_{sys}				$\delta\beta/\beta$ (linear)			
	$f_2 = 0.05$	0.10	0.25	0.50	0.05	0.10	0.25	0.50
15°	30.3	60.9	154	312	+1.5%	+3.0%	+7.9%	+17%
30°	116	234	605	1281	+6.0%	+12%	+35%	+89%
60°	266	546	1481	3465	+11%	+25%	+83%	+308%
90°	446	912	2442	5559	-8.4%	-17%	-39%	-70%

All 16 Not identifiable. These are linear projections: beyond $d_{\text{sys}} \sim 1$ the parameter biases are qualitative (V1). The scatter grows too: under the directional sea $\sigma_{\ln \beta}$ is 2.0 to 3.4 times nominal. Pre-amendment: d_{sys} 7.5 to 1342.

Even 5% secondary energy breaks the inverse, and the damage grows with separation. At 90° a large share of the bias goes into R ($\delta R/R = -3.3\%$ against $\delta\beta/\beta = -8.4\%$ at 5%). The stiffness bias per unit energy is largest at 60°, while the Fisher distance is largest at 90°.

3.2 Small contamination: the tolerances

The bias is linear in f_2 . d_{sys}/f_2 is constant to 0.04% over $f_2 = 2.5 \times 10^{-4}$ to 10^{-3} , with slopes 605, 2295, 5197 and 8738 for 15, 30, 60 and 90° (3.1 to 4.0 times the pre-amendment slopes). It is still within 2% at $f_2 = 0.05$. The crossings therefore lie in a genuinely local regime.

ϕ	f_2 at $d_{\text{sys}} = 0.5$	$d_{\text{sys}} = 1$	$ \delta\beta/\beta = 1\%$	f_{detect} (coherence)	early warning
15°	0.082%	0.165%	3.32%	0.118%	no ($\times 0.70$)
30°	0.0218%	0.0435%	0.869%	0.0082%	yes ($\times 2.6$)
60°	0.0096%	0.0192%	0.467%	0.00092%	yes ($\times 10.5$)
90°	0.0057%	0.0115%	0.591%	$\leq 0.0005\%$ (below floor)	yes ($> \times 11$)

pre-amendment: $d_{\text{sys}} = 0.5$ at 0.332, 0.086, 0.030, 0.022%; f_{detect} 2.01, 0.134, 0.0129, 0.0063%; warning at 60 and 90° only

Fractions of the total incident energy. Early warning: $f_{\text{detect}} < f_2(d_{\text{sys}} = 0.5)$, factor = $f_2(d_{\text{sys}} = 0.5)/f_{\text{detect}}$. At 90° detection is already met at the search floor $f_2 = 5 \times 10^{-6}$.

Statistical inconsistency arrives long before a large physical error: the 1% stiffness criterion needs 40 (15°, 30°), 49 (60°) and 103 (90°) times the energy that gives $d_{\text{sys}} = 0.5$. The physical tolerance moved the other way from the significance tolerance. It rose at 15 to 60° (3.32% against 1.88% at 15°) and fell at 90° (0.59% against 1.75%), because the amendment reweighted the bins and so changed how the mixture projects onto β and R .

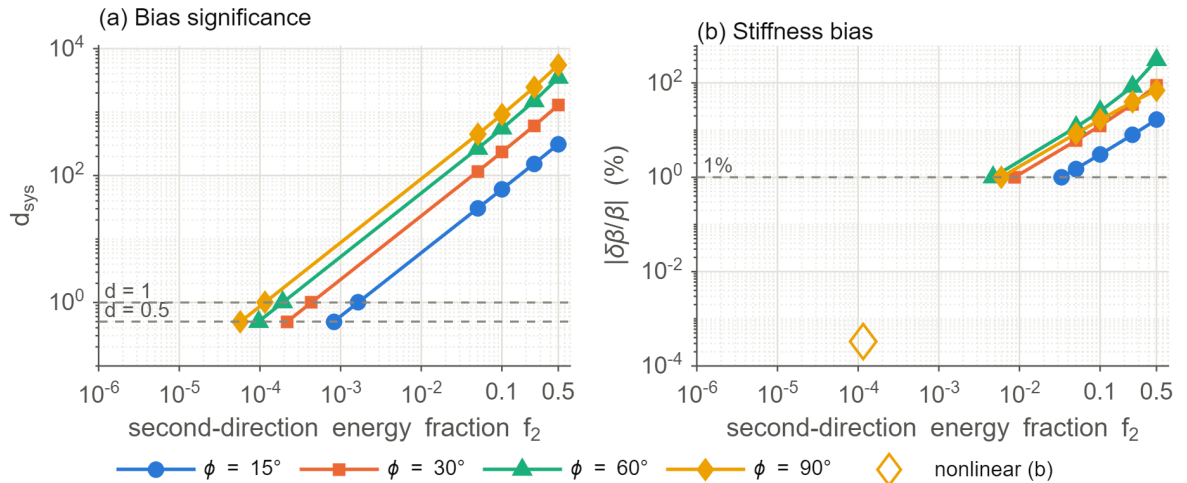


Figure 1. Inverse bias against the second-direction energy fraction, per separation ϕ : (a) d_{sys} with the 0.5 and 1 limits; (b) $|\delta\beta/\beta|$ with the 1% limit. Registered points and the bisected crossings. Open marker: the nonlinear inversion at the 90°, $d_{\text{sys}} = 1$ crossing. It shows the *total* stiffness shift (Welch plus directional), which nearly cancels there, hence its position near $10^{-4}\%$.

3.3 Nonlinear check

At $\phi = 90^\circ$, $f_2 = 1.149 \times 10^{-4}$ ($d_{\text{sys}} = 1.004$), the G3 solver converged in 2 iterations (8 model evaluations, 254 s). It matched the linear prediction $\theta_{\text{true}} + \delta\theta_W + \delta\theta_{\text{dir}}$ exactly (gap 0.000σ): $\delta R/R = -0.0041\%$, and $\delta\beta/\beta \approx 0$. The

stiffness shift is nearly zero because the directional part (about -0.019% by linear scaling) and the Welch part almost cancel at this point; the agreement is what matters. The tolerances are therefore not a linearisation artefact, and the failure is not an optimiser pathology: the solver finds the wrong answer cleanly.

3.4 Coherence as a validity diagnostic

The single-direction pair coherence is now 0.9988 to 0.9989 over the band (about 0.99 before the amendment). The measured null scatter of the band-mean Fisher z was $[0.0095, 0.0099, 0.0095]$, against 0.0101 for independent bins (variance ratios 0.89, 0.97 and 0.88; effective bins 86 to 95 of 84). The independent-bin value is therefore slightly conservative here; the calibrated values are used.

Coherence now warns first at 30° , 60° and 90° , and fails narrowly at 15° . Close headings give nearly collinear response vectors, so the spectral matrix stays nearly rank one while the very precise inverse is already biased. Detection improved about fifteenfold at every angle (2.01% to 0.118% at 15°), more than the significance tolerance tightened (fourfold). With the noise floor relatively lower, a small rank-two component is easier to see.

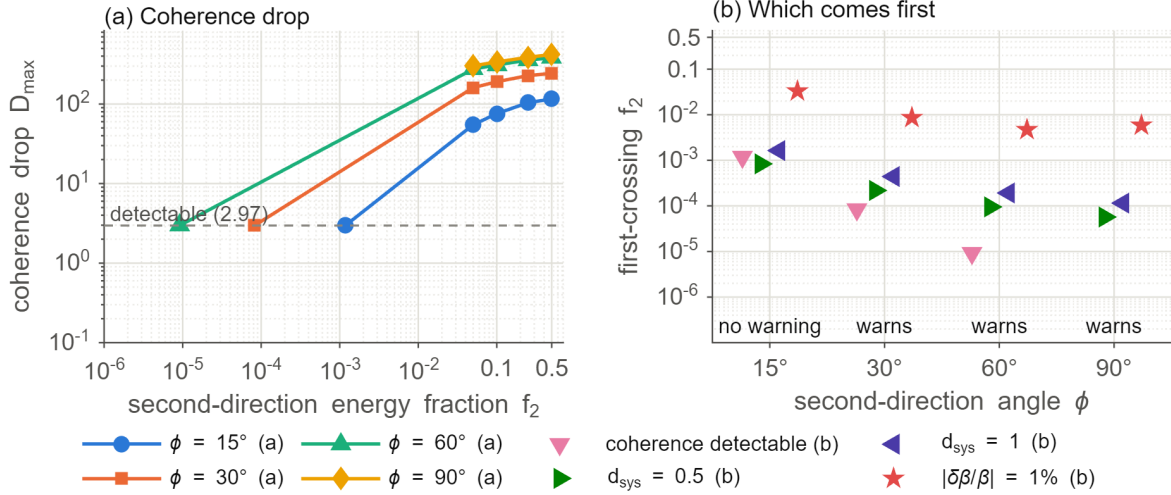


Figure 2. (a) Expected standardised coherence drop $D_{\max}$ against f_2 , with the 80%-power detectability threshold. (b) First-crossing fractions per separation: coherence warns early where its marker lies below the $d_{\text{sys}} = 0.5$ marker. At 90° the detection marker is absent because detection lies below the search floor ($f_2 \leq 5 \times 10^{-6}$).

Noise-floor sensitivity (post-run diagnostic, not registered). The baseline coherence is set mostly by the noise floor, so an error ϵ in the assumed S_{nn} shifts it in Fisher z and can fake or hide a drop.

ϕ	f_{detect} with the baseline at $S_{nn}(1 + \epsilon)$					$f_2(d_{\text{sys}} = 0.5)$
	$\epsilon = -10\%$	-5%	0	$+5\%$	$+10\%$	
15°	false alarm	0.071% (early)	0.118% (late)	0.165% (late)	0.213% (late)	0.082%
30°	false alarm	0.0049% (early)	0.0082% (early)	0.0115% (early)	0.0148% (early)	0.0218%
60°	false alarm	0.0005% (early)	0.0009% (early)	0.0013% (early)	0.0017% (early)	0.0096%
90°	false alarm	0.0002% (early)	0.0005% (early)	0.0007% (early)	0.0009% (early)	0.0057%

With no second direction, an underestimated noise floor alone reaches the 5% rejection threshold at $\epsilon = -5.82\%$ and the 80%-power threshold at -8.06% (pre-amendment -4.39% and -6.07%). An overestimate masks real drops ($D_{\max} = -2.2$ at $+10\%$).

For separations of 30° and more, the early warning now survives a noise-floor error of -5 to $+10\%$. Before the amendment it failed at $+5\%$ for 30° and at $+10\%$ for 60° . The requirement is still much stricter than the inverse itself needs (R3: no noise-misspecification tolerance is reached for the LSM6DSV16X), and asymmetric: underestimating S_{nn} by more than about 6% manufactures directional detections.

4 Interpretation

- The failure is structural.** Under independent directions $S_{jr}/S_{rr} \neq H_j/H_r$. The frozen inverse converges and returns biased (β, R); no estimator tuning removes this.
- Tiny secondary energy is statistically decisive**, because the nominal inverse is so precise ($\sigma_{\ln \beta} = 0.050\%$): 0.006 to 0.08% of the incident energy reaches $d_{\text{sys}} = 0.5$. This is the same pattern as V1: once precision is high, any model mismatch is visible.
- Fisher significance and physical error answer different questions.** d_{sys} flags the mismatch 40 to 100 times earlier than the 1% stiffness criterion. In physical terms, a secondary direction carrying under 0.5% of the energy keeps β within 1% at every tested separation.
- Coherence is a usable validity diagnostic for separations of 30° and more.** It warns 2.6 to more than 11 times before the bias becomes significant, and stays reliable for a noise-floor error of -5 to $+10\%$. It cannot certify the single-direction assumption for close headings (15°), where the bias arrives first, though only narrowly.

5. **This motivates D3.** V2 identifies the failure at the Level 2C layout; D3 asks whether sensor count and placement can make the inverse robust to it.

Conclusion (thesis wording). V2 deliberately violated the single-input assumption by adding a second independent incident direction while the inverse remained frozen to a single 0° input. The resulting transmissibility mismatch was extremely influential: the $d_{\text{sys}} = 0.5$ tolerance fell from about 0.08% secondary energy at 15° separation to 0.006% at 90° , while a 1% stiffness bias required 0.5 to 3.3%. These small-fraction crossings lie in a strictly linear regime, and a full nonlinear inversion at the most sensitive $d_{\text{sys}} = 1$ crossing agreed exactly with the local prediction. Coherence gave an early warning at separations of 30° and more, robust to a noise-floor error of -5 to $+10\%$, but not at 15° . Coherence is therefore a useful diagnostic for separated independent forcing, but cannot by itself certify the single-direction model.

Limitations.

- One floe (p_0), the Level 2C layout, one sea shape and four separations.
- Two ideal, fully independent components; a real sea is spread and partly coherent (D3b adds a spread case).
- f_2 is an incident-energy fraction, while the effective mixing is weighted by the reference response.
- The class uses the nominal σ (the frozen inverse's ruler); the actual scatter under the directional sea is 2 to 3.4 times larger.
- The coherence baseline is the model prediction at the true parameters; the noise diagnostic is post-run.
- Apart from the forcing, truth and inverse share the EMM, including its truncation (V1).

Requirement for the experiment and the thesis

The single-direction transmissibility inverse must not be applied blindly under multidirectional forcing. Secondary energy of order 0.01% of the total biases it beyond its own uncertainty, and 0.5 to 3% biases the stiffness by 1%. For secondary directions 30° or more away, coherence serves as an early warning provided the in-band noise PSD is calibrated to within about -5 to $+10\%$ (underestimation creates false alarms). Closely separated components can bias the inverse before any coherence warning. Directional forcing must therefore be modelled explicitly or treated as a validity condition of the frozen inverse.

Output (Results/V2/)	Contents
V2_20261009_133308.mat / _summary.txt	results and config; headline numbers
V2_grid.csv, V2_crossings.csv, V2_nonlinear.csv	16 cases; first crossings; nonlinear check
diagnostics/V2noise_20261009_134221.mat, V2_noise_*.csv	noise-floor diagnostic
figures/V2_fig1, V2_fig2 (.pdf/.png)	Figures 1 and 2 (plotDirectionalForcingStudy)
Superseded	V2_20261005_142206, V2noise_20261005_150224 (pre-amendment)